Pratik Pujari

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Education

University of Colorado Boulder — Boulder, CO
Master of Science in Computer Science | GPA: 3.8/4.0 — *Aug. 2025 – Present*

Relevant Coursework: Algorithms, Object-Oriented Design, Linux System Administration, Deep Learning, NLP, Graphics

Sardar Patel Institute of Technology — Mumbai, India
Bachelor of Technology in Computer Engineering | GPA: 3.96/4.0 — *Oct. 2020 – May 2024*

Experience

Software Engineer Intern — May 2026 – Aug 2026
Visa Inc. — *Highlands Ranch, CO*

- Architected **VisionX**, an AI log-triage platform for ATM monitoring: Drain3 templating, a 20-class embedding classifier (**82% F1**) with **LLM** escalation, and HDBSCAN clustering condensing ∼**30K** templates into ∼**200** ranked issues.
- Built a **LangGraph** remediation agent that automatically picks up P0 issues, clones the failing repository, diagnoses the root cause, and ships each fix as a **GitHub** pull request with a **Jira** comment; its fixes for real issues were merged.
- Engineered a fault-tolerant **FastAPI** + **Kafka** pipeline (9 topics, retry backoff, dead-letter queue) over **PostgreSQL**, sustaining ∼**1M logs/hour** by classifying each log template once and caching embeddings in **Redis** and SQLite.
- Cut triage time **50–60%** versus manual **Splunk** and **Grafana** searches; demoed VisionX live to **VP-level** leadership.
- Stopped false on-call pages by keeping backfilled logs (600s+ lag) out of live **Prometheus** and **Alertmanager** rules.
- Built a **0–100** outage risk engine on seasonal baselines, flagging at-risk ATM sources early with a recommended action.

Software Developer Intern — Nov. 2023 – Dec. 2024
Aim4U Software Solutions Private Limited — *Mumbai, India*

- Architected the cloud backend of **Dermalens**, an AI telehealth platform: a **Django** REST API and website for booking, e-prescriptions, and **Stripe** payments with webhook-confirmed bookings, refunds, and **Connect** doctor payouts.
- Replaced the legacy TensorFlow predictor with a **Detectron2** + **PyTorch** skin-disease model (**90–95%** accuracy), served as serverless inference on **AWS Lambda** behind **API Gateway** so it scales independently of the backend.
- Deployed the backend on self-managed **Kubernetes** on **EC2**; owned **Route 53**, **S3**, **ECR**, **IAM**, and **CloudWatch**.
- Pitched Dermalens to investors at 3+ startup competitions; shipping the judges' feedback lifted downloads by **23%**.
- Built **GitHub Actions** CI/CD with lint, pre-merge checks, and integration tests; mentored **4–5 junior interns**.

Research and Software Intern — Jan. 2023 – June 2023
Acuradyne Systems Private Limited, IIT Bombay Research Park — *Mumbai, India*

- Co-designed a chunked **BLE** protocol on **ESP32** for an IoT pulse monitor: **566%** faster transfer (**850ms to 127ms**).
- Migrated patient records from offline, device-locked **SQLite** to **Firestore**, syncing the same data across all devices.
- Built a **Node.js** backend on **GCP** with **JWT** auth, generating **1K+** clinical pulse reports per day via **Puppeteer**.
- Replaced manual USB flashing with **OTA** firmware updates (checksums, rollback), cutting production time ∼**40%**.
- Automated tester builds for end-to-end internal testing and set up a **Docker** dev environment; led **3–4 interns**.

Projects

Pivot – Agentic RL Research | *Rust, Python, Firecracker, MLX* — May 2026 – Present

- Built **Pivot**, a reinforcement learning system that assigns step-level credit to coding agents by forking their sandboxes mid-task and measuring how each action changes the chance of success, instead of a single reward per task.
- Engineered a **Rust** fork runtime for **Firecracker** microVMs with userfaultfd paging, huge pages, and copy-on-write disks, cloning a live VM **16** ways in **262ms** and restoring state **13x** faster than **Docker** replay (208ms vs. 2.7s).
- Cut 16-clone parallel test runs from **53s** to **715ms** (**74x**) via on-demand memory paging and idle-VM pausing.
- Improved step-credit accuracy **2x** over **GRPO** and **4x** over a frozen 3B **LLM** critic against a brute-force oracle.
- Raised a 1.5B coding model's held-out success to **62%** vs. **27%** for **GRPO** at equal compute, using **LoRA** fine-tuning.

Echo – Chat App | *Go, Kafka, PostgreSQL, React, Docker* | Live | GitHub — Mar. 2025 – Sept. 2025

- Shipped **Echo**, a chat and video-call app of **8 Go microservices** and **React**, running in ∼**500 MB** of RAM.
- Built horizontally scalable **WebSockets** on **Redis** pub/sub, holding **10K+** live connections at **sub-10ms** latency.
- Engineered gap-free, duplicate-free delivery with **Kafka**, per-chat sequence numbers, and **Redis** idempotency keys.
- Hardened production with a **Cloudflare Tunnel** (zero open ports), health checks, and auto-rollback of failed deploys.

Technical Skills

Languages & AI: Python, Go, Java, TypeScript, Kotlin, C++, SQL, Rust, PyTorch, LangGraph
Backend & Systems: Distributed Systems, Microservices, Kafka, gRPC, WebSockets, FastAPI, Django, React
Cloud & Data: AWS, GCP, Kubernetes, Docker, PostgreSQL, Redis, Prometheus, Grafana, Linux